

The Grammar of Absence: A Relational Approach to Abessive Constructions

Abstract

While comitative and instrumental relations have received extensive attention in the typological and generative literature, their negative counterpart—the caritive or abessive construction (e.g., *without*, *sans*, *senza*)—remains structurally under-theorized. Descriptive typologies cross-linguistically observe that expressions of absence are systematically more complex and heavily marked than expressions of presence (Stolz et al. 2007). This paper proposes a formal explanation for this markedness asymmetry based on a Double-Relator Model (DRM) and mereotopology. We argue that while comitative/instrumental markers are primitive relational heads computing simple inverse inclusion ($\supseteq$), abessive markers are Axial Parts (AxPs) or Relational Nouns. Drawing on Varzi's (2007) mereotopology, we show that grammaticalizing an "absence" cognitively amounts to creating a topological *void*, an operation that requires projecting a discrete spatial/abstract boundary (an AxP) to grant "ontological dignity" to the empty region. Integrating semantic insights from Matushansky (2019) on kind-denoting remnants and Phillips (2022) on negative existentials, we demonstrate that caritive constructions uniformly project a bipartite syntactic structure, elegantly accounting for the morphosyntactic behavior of caritives across Indo-European and non-Indo-European languages.

Keywords: Caritive, Comitative-abessive asymmetry, Double-Relator Model, Axial Parts, Mereotopology.

1. Introduction: The Asymmetry of Presence and Absence

The grammatical encoding of accompaniment (comitative) and tool-use (instrumental) is widely attested and has been systematically analyzed in both the generative and functionalist literature. Conversely, the logical negation of these categories — the *abessive*, *privative*, or *caritive* construction (e.g., English *without*, French *sans*, Italian *senza*) — has been treated as a peripheral phenomenon in the literature. When discussed, caritive expressions are routinely conceptualized merely as the negated counterparts of comitatives or instrumentals, forming a simple functional opposition of categories.

However, a closer look at cross-linguistic data reveals a pervasive and systematic structural asymmetry: caritive and abessive are typologically much more heavily marked than their positive counterparts. Kiparsky (2012) identifies a fundamental typological implication: if a language has a ‘without’ case, it also has a ‘with’ case, but not necessarily conversely.

Furthermore, as thoroughly documented by Stolz et al. (2007), an expression meaning ‘without’ can frequently be morphologically derived from an expression meaning ‘with’ (usually *via* the agglutination of a certain morpheme). Consider the contrast in Hixkaryana (Cariban) in (1), where the caritive is simply the comitative suffix bound to a ‘negation’ morpheme:

- (1) Hixkaryana (Stolz et al. 2007)
- a. kamo ro-akoro
sun 1SG-with
‘the sun with me’ (Comitative)
 - b. kamo ro-akoro-hra
sun 1SG-with-NEG
‘the sun without me’ (Caritive)

When languages possess dedicated, non-agglutinative expressions for the abessive, these markers almost universally exhibit higher segmental and morphological complexity than the comitative. They display a striking resistance to becoming bound morphemes, overwhelmingly preferring the status of adpositions / complex particles. This asymmetry is fully transparent in Romance languages like Italian. While the comitative is expressed by a simple, monosyllabic preposition (2a), the caritive inherently requires a heavier, polysyllabic

marker, which frequently forces the projection of an additional, lower relational element (*di* 'of') to introduce specific Grounds, such as pronouns (2b):

(2) Italian

- a. Maria è uscita con te.
 Maria be.3SG gone.out with you
 'Maria went out with you.'
- b. Maria è uscita senza (di) te.
 Maria be.3SG gone.out without of you
 'Maria went out without you.'

This fact raises a fundamental question at the syntax-semantics interface: Why does the grammar of absence systematically require more 'structure' than the grammar of presence? In this paper, we propose that this morphosyntactic markedness is not an accidental typological tendency, but a direct reflex of the mereotopological nature of absence. By adopting a strictly phrasal and relational approach based on a Double-Relator Model (DRM) (Franco 2024; Franco & Schirato 2026), we argue that grammaticalizing an "absence" amounts to creating a topological *void*. While comitative/instrumental markers are primitive relational heads computing simple zonal inclusion, caritive markers must universally project as complex spatial or relational nominals — namely, Axial Parts (AxPs) (Svenonius 2006). Drawing on Varzi's (2007) mereotopology, we show that to identify a void as a computable entity, the syntax is forced to draw a discrete boundary (an AxP) to grant "ontological dignity" to the empty region itself, requiring more structural complexity. Integrating semantic insights on kind-denoting remnants (Matushansky 2019) and negative existentials (Phillips 2022), we demonstrate that caritive constructions uniformly project a bipartite syntactic structure.

2. Theoretical Background: Relational Primitives and Axial Parts

A proper understanding of the syntactic divide between presence and absence requires a framework that decomposes spatial and abstract relations into minimal units. In this section, we introduce the theoretical prerequisites of our analysis, integrating the syntax of elementary predicates with the geometry of spatial nominals.

2.1 Elementary Predicates of Inclusion

Recent literature demonstrates that oblique cases and adpositions (Manzini & Savoia 2011, Manzini & Savoia 2018) such as the genitive *of* (*di* in Italian), act as elementary predicates expressing "zonal inclusion", notated as ($\subseteq$), a concept originally coined by Belvin & den Dikken (1997). In a standard possessive configuration, such as the Italian *i bambini della mamma* ('the children of the mother'), the genitive preposition *di* embeds the possessor (*la mamma*), indicating that the possessum (*i bambini*) is structurally included within the possessor's domain. However, this mereological relation can be morpho-syntactically reversed by utilizing the preposition *with* (cf. Levinson 2011). In configurations like *la mamma con i bambini* ('the mother with the children'), the comitative preposition *con* ('with') embeds the possessum rather than the possessor. This yields a relation of reverse inclusion ($\supseteq$), where the external argument of the adposition (the whole) includes its complement (the part). It is precisely upon this ($\supseteq$) architecture of the comitative/instrumental that the grammar of the abessive ('without') builds its spatial and abstract boundary of zonal exclusion. Thus, we assume that adpositions and oblique cases are not semantically vacuous functional heads inserted merely for Case-theoretic reasons. Instead, they are active, interpretable elementary predicates that establish part-whole relationships. The relation of inverse inclusion ($\supseteq$) is particularly relevant for the grammar of accompaniment and instrumentality. In a standard comitative construction (e.g., Italian *con i bambini* 'with the children'), the preposition *con* lexicalizes a ($\supseteq$) predicate (cf. Franco & Manzini 2017).¹

- (3) Italian
[La mamma] [$\supseteq$ con [i bambini]]
'The mother with the children'

Cognitively and syntactically, (3) maps onto a space of simple accompaniment: the Figure (the mother) "includes" or is strictly concomitant with the Ground (the children). Because this

¹ A structural distinction between entity-modification and event-modification is explicitly formalized for positive *with*-phrases in Franco & Manzini (2017). While adnominal possession as in (3) establish an elementary relation of inverse inclusion ($\supseteq$) strictly between two nominal entities (e.g., *the mother with the children*, where the higher DP mereologically includes the lower DP), instrumentals attach to the verbal spine. In an instrumental configuration such as *Ho tagliato il pane con il coltello* ('I cut the bread with the knife'), the relational head ($\supseteq$) takes the inanimate DP (*il coltello*) as its internal argument, while its external argument is the entire VP event (*tagliare il pane*). The syntax dictates that the caused event of cutting the bread mereologically includes the knife.

operation merely links two positively integrated entities, it requires minimal syntactic effort and may be universally lexicalized by primitive, monosyllabic relators (or simple case affixes).

2.2 Axial Parts and the Double-Relator Model

While simple prepositions map direct relational predicates, natural language frequently resorts to a more complex architecture to define specific regions of space or abstract domains. Drawing on the seminal work of Svenonius (2006), we recognize the existence of Axial Parts (AxPs). Axial Parts (e.g., English *front*, *top*; Italian *fronte*, *capo*, *cima*) are elements of nominal origin that have been stripped of their standard referential architecture (e.g., they lack D-layers and do not pluralize) to function as spatial or relational positioners. They do not denote a referential object, but rather a specific region or boundary defined in relation to a Ground. Franco (2024) and Franco & Schirato (2026) systematize the interaction between elementary predicates and Axial Parts through a Double-Relator Model (DRM). Within the DRM, what traditional grammars label as "complex prepositions" are analyzed as fully transparent, bipartite syntactic configurations. An Axial Part is sandwiched between two relational heads, yielding a structure like the one in (4)

(4) [Figure [p_upper $\subseteq$ [AxP [p_lower $\subseteq$ Ground]]]]

In the structure illustrated in (4), p_upper functions as the higher relator linking the external Figure (conceived as a modified DP or event) to the Axial Part. Conversely, p_lower operates as a lower relator, establishing the relation between the Axial Part and the Ground DP (or event). Each instance of p corresponds to an abstract relational predicate—namely, a minimal prepositional unit expressing a generalized inclusion relation, which can be represented as ($\subseteq$). This configuration captures the compositional rationale underlying the Double-Relator Model: the Figure is contained within the region specified by the Axial Part, and this region is itself contained within the Ground. Put differently, the structure encodes a layered inclusion relation (Figure $\subseteq$ Region $\subseteq$ Ground), thereby motivating the postulation of two distinct relational heads.

This layered architecture is overtly visible in the diachrony of Romance languages. As demonstrated by Franco & Schirato (2026), Old Italian systematically deployed two overt relators to embed an AxP, as shown in (5).

(5) Old Italian (from Franco & Schirato 2026)

- a. ...in cima de la torre
 in top of the tower
 '...on top of the tower.' (Jacopo da Firenze, *Tractatus algorismi*, 1307)
- b. [p_upper in [AxP cima [p_lower de [DP la torre]]]]

In (5) the lower relator *de* 'of') establishes a basic inclusion relation ($\subseteq$) between the Ground and the AxP (effectively marking the Ground as the "possessor" of the region). The upper relator (*in* 'in') links the newly formed axial region to a Figure (a DP or an event). Crucially, as a result of grammaticalization (Roberts & Roussou 2003), the upper and lower relators can frequently undergo phonological reduction or remain silent ($\emptyset$) in modern varieties, yet their syntactic and semantic licensing roles remain fully active.

2.3 Exceptives and Caritives: From Grammar of Exclusion to the Grammar of Absence

The DRM has recently been applied to resolve the syntactic puzzle of exceptive constructions (e.g., *everyone except John*). Exceptive markers (see Von Stechow 1993, Moltmann 1995) are not clausal complementizers or Boolean operators involving clausal ellipsis, as argued in recent literature (see Vostrikova 2019, 2021, and references cited there) but rather grammaticalized Axial Parts, within a layered PP skeleton.

In a quantificational domain, subtracting an entity from a universal set ($W \setminus P$) requires the definition of an exclusion boundary. It is possible to assume that the exceptive marker (e.g., Italian *tranne, eccetto, ad eccezione di*) are AxPs that create this boundary, taking the subtracted entity as its internal Ground, with the relators sometimes realized as null elements ($\emptyset$), following insight by Kayne (2005). Consider the data below, showing a Relation Noun (6a) and AxP (6c), introducing an exceptive relation. We assume that the underlying syntax is the same: the only difference is that the inclusion relators may be unpronounced when the exceptive marker surfaces as an AxP.²

² When dealing with highly grammaticalized, monomorphemic exceptive markers like *tranne* ('except') in (5b), the relators are not phonologically realized. From a theoretical standpoint, the absence of overt adpositions in these configurations can be formalized in two ways. One may postulate the existence of phonologically silent relators ($\emptyset$) projecting their own functional heads, along the lines of Kayne (2005). Alternatively, adopting a Nanosyntactic perspective (Starke 2009; Caha 2009), the intense grammaticalization of these markers can be modeled as a process of *phrasal spell-out*. Under this view, a single, heavily grammaticalized lexical item like *tranne* is capable of spelling out the entire complex syntactic spine spanning from the lower to the upper relator, absorbing their structural features.

(6) Italian

- a. Tutti, ad eccezione di Gianni, sono partiti
'Everyone except Gianni left.'
- b. [p_upper ad [AxP eccezione [p_lower di [DP Gianni]]]]
- c. Tutti, tranne Gianni, sono partiti.
All except Gianni be.3PL left
'Everyone except Gianni left.'
- d. [p_upper Ø [AxP tranne [p_lower Ø [DP Gianni]]]]

The ontological step from exceptives to caritives is minimal. Exceptives execute a *set subtraction* operating on quantifiers. Caritives, on the other hand, operate on what Belvin & den Dikken (1997) call "zonal inclusion"— the possessive or eventive zone associated with a subject or an event. To state that someone is "without" something is to assert a *zonal exclusion*. Establishing an exclusion zone, whether quantificational (exceptives) or possessive/eventive (caritives), inherently demands the complex, bipartite syntactic architecture of an Axial Part rather than a simple preposition/case marker. As defined by Oskolskaya et al. (2020), the caritive describes the non-involvement or absence of a participant (the absentee) from a situation, often parasitic on an underlying possessive relation (Matushansky 2019). Thus, while exceptives execute *set subtraction*, caritives identify *zonal absence*. Yet, as we will see, both operations trigger the exact same morphosyntactic repair mechanism: the projection of an Axial Part.

2.4 The Phrasal Nature of Caritives: Zonal Absence vs. Clausal Negation

A crucial piece of evidence supporting the Double-Relator Model over a logical negation approach comes from the grammatical encoding of absence. Typologically, caritive (or abessive) constructions are frequently analyzed as the negative counterparts to comitatives and instrumentals (Stolz et al. 2007). Under a clausal reduction hypothesis, one might be tempted to treat a caritive marker like *senza* ('without') as a mere surface reflex of clausal coordination involving logical negation (e.g., deriving *without X* from an elided *but not [verb] with X*).

However, making the allegedly elided clausal structure explicit reveals that caritive constructions operate strictly at the phrasal level via mereological subtraction ("zonal absence"), generating semantic and pragmatic outputs that diverge sharply from true clausal boolean coordination. Consider the minimal pairs in (7) and (8).

(7) Italian

- a. Gianni ha tagliato il pane senza il coltello.
 Gianni has cut the bread without the knife
 'Gianni cut the bread without the knife.'
- b. Gianni ha tagliato il pane, (ma) non ~~[ha tagliato il pane]~~ con
 Gianni has cut the bread but not ~~[has cut the bread]~~ with
 il coltello...
 the knife...
 'Gianni cut the bread, but he didn't [cut the bread] with the knife....'

While (7a) simply denotes a cutting event characterized by the absence of a specific instrument, the clausal coordination in (7b) triggers a completely different pragmatic effect. The logical negation 'non' operating over the fully projected (yet elided) clausal node inherently activates a set of Focus alternatives (Rooth 1992). It negates the proposition that a knife was the instrument of the cutting event, strongly implicating that an alternative instrument was employed (e.g., *his hands, a saw, etc.*). The caritive Axial Part *senza* in (7a) does not trigger such contrastive clausal alternatives; it directly modifies the event by structurally projecting a boundary of exclusion. The inadequacy of the clausal/boolean analysis becomes even more glaring when examining inalienable part-whole relations, as shown in (8):

(8) Italian

- a. Ho visto una ragazza senza un orecchio.
 have.1SG seen a girl without an ear
 'I saw a girl without an ear.'
- b. #Ho visto una ragazza, (ma) non ~~[ho visto una~~
 have.1SG seen a girl but not ~~[have.1SG seen a~~
~~ragazza]~~ con un orecchio.
~~girl]~~ with an ear
 '#I saw a girl, but I didn't [see a girl] with an ear.'

Sentence (8a) is perfectly felicitous: the caritive marker acts as a relational head operating over the nominal domain, mereologically subtracting one specific inalienable part (an ear)

from the whole (the girl). Conversely, the clausal coordination in (8b) is semantically anomalous. Because *ma non* operates as a boolean operator over a hidden propositional node, it applies logical negation to the proposition *I saw a girl with one ear*. Pragmatically, this yields the bizarre reading that the speaker saw a girl, but the proposition "she had exactly ONE ear" is false, thus implicating, among other alternatives, that she possessed the standard biological amount (two). The clausal derivation completely fails to capture the privative/subtractive meaning.

These contrasts prove that caritive constructions cannot be collapsed into clausal coordination with logical negation. Instead, just like true exceptives, caritives require the syntax of a complex phrasal constituent based on the Double-Relator Model. In plain words, a marker like *senza* ('without') does not introduce a hidden negative clause; it functions as an Axial Part—a relational element denoting an "empty region" or "zonal absence".

Syntactically, this Axial Part is simply sandwiched between two elementary relators. The higher relator links the main event (e.g., *cutting the bread*) to this zone of absence, while the lower relator connects the zone of absence to the nominal Ground (*the knife*). The structural reality of this lower relator is fully transparent in Italian when *senza* takes a pronominal Ground, as the relator overtly surfaces as the genitive preposition *di* (e.g., *senza di me*, 'without of me'). Ultimately, the syntax merely states that an event (or an entity) is located within the "absence" of a specific object, computing mereological subtraction directly at the phrasal level without any need for invisible clausal nodes. Another argument against a clausal negation approach to caritives (i.e., treating *without* as derived from an elided *but not with*) comes from cliticization patterns in Italian. Within the Double-Relator Model, the caritive marker *senza* ('without') is an Axial Part that structurally requires a lower relator (*p lower*) to introduce the Ground. As independently established, when the Ground of an AxP is pronominalized in Italian, this lower relator explicitly surfaces as the genitive preposition *di* ('of') (e.g., *senza di me* 'without of me'). If this phrasal, relational architecture is correct, we predict that the Ground of *senza* should be able to undergo cliticization using *ne*—the canonical Italian clitic strictly reserved for pronominalizing genitive/ablative PPs headed by *di* or *da*. This prediction is robustly borne out, as shown by the minimal pair in (9) and (10):

(9) Italian

- a. Non posso fare a meno della pasta.
 Not can.1SG do to less of.the pasta
 'I cannot do without pasta.'

- b. Non posso far-ne a meno.
 Not can.1SG do-of.it to less
 'I cannot do without it.'

(10) Italian

- a. Non posso stare senza la pasta.
 not can.1SG stay without the pasta
 'I cannot stay without pasta.'
- b. Non posso star-ne senza.
 not can.1SG stay-of.it without
 'I cannot stay without it.'

The complex adpositional locution *a meno di* in (9), in our view, is a transparently analytic Axial Part: it consists of a relational noun (*meno*) sandwiched between an upper relator (*a*) and a lower genitive relator (*di*). Consequently, its Ground is perfectly cliticized by the genitive *ne* (9b). Strikingly, the grammaticalized marker *senza* in (10) allows the exact same genitive cliticization (10b).³ This constitutes compelling evidence that *senza* projects the exact same underlying relational structure, containing a (silent) genitive *p lower* that licenses the *ne* clitic.⁴

Crucially, this cliticization pattern is utterly derivable under a clausal negation hypothesis. If *senza la pasta* were syntactically derived from a negated comitative preposition (*non con la pasta* 'not with the pasta'), the emergence of the genitive clitic *ne* would be syntactically impossible. In Italian, the comitative preposition *con* ('with') strictly forbids *ne* cliticization (**[stare con la pasta] → *[starne con]*). The robust availability of the *starne senza* pattern

³ It should be noted that in standard prescriptive Italian, the cliticization of the complement of *senza* with *ne* (e.g., *starne senza*, *farne senza*) is sometimes marked as colloquial or restricted to Northern regional varieties, with the standard language preferring the analytic Axial Part *farne a meno*. However, from a syntactic perspective, the fact that the Italian diasystem generates and allows this cliticization is theoretically highly significant: it confirms the underlying structural availability of a genitive lower relator (*p lower*) within the argument grid of the caritive marker.

⁴ One might wonder how the genitive clitic *ne* can be extracted if *senza* does not overtly assign genitive case to a full DP Ground (e.g., *senza la pasta* rather than **senza della pasta*). This surface mismatch is perfectly predictable within the Double-Relator Model. As we have said, grammaticalized Axial Parts like *senza* feature a phonologically null lower relator (*p lower* = $\emptyset$) when selecting a full DP. Crucially, however, when the Ground is pronominalized, this lower relator must overtly surface as the genitive preposition *di* (e.g., *senza di me* 'without of me'). Since *ne* is a pronominal clitic, its extraction naturally activates this latent genitive valency, confirming the (covert) presence of the lower relator.

therefore proves that caritive markers are true relational heads assigning genitive case to their nominal Ground.

3. The Mereotopology of Voids and the Double-Relator Model

To capture the syntax of caritives, we adopt the Double-Relator Model (DRM) just outlined. The Comitative relation expresses straightforward inverse inclusion ($\supseteq$). Cognitively, it maps onto a space of simple accompaniment and is headed by a primitive, monosyllabic relator, as we have highlighted in (3). The syntax effortlessly links two existing, integrated entities, as shown in Italian (11a).

(11) Italian

- a. La mamma con te /*con di te
 the mother with you/with of you
 'The mother with you.' (Inverse Inclusion $\supseteq$)
- b. La mamma senza (di) te
 the mother without (of) you
 'The mother without you.' (Zonal Exclusion)

Conversely, the Caritive relation in (9b) asserts an absence. A potential criticism of our approach, as we have already pointed out in Section 2.4, might be that absence could simply be derived by merging a standard negative functional projection over a preposition (e.g., [NegP not [PP with you]]). Actually, we are showing with Italian data, that languages with dedicated caritive markers do not project a negative operator over a simple preposition. Instead, they grammaticalize complex markers. To understand why, we must look at the mereotopology of absence.

In mereotopological terms (Varzi 2007), establishing an absence creates a "hole" or "void" within a region. Pure mereology—the strict formal theory of parthood—only possesses the conceptual resources to deal with existing matter. It lacks the mathematical and cognitive tools to account for voids and disconnectedness, because a hole is, essentially, "nothingness." As Varzi (2007) points out, one cannot directly compute or isolate a void; to identify a void as

a discrete entity, one must trace its "immaterial boundaries", an operation that confers "ontological dignity" to the hole itself.⁵

We argue that, syntactically, the grammar perfectly mirrors this topological necessity. An elementary prepositional relator is incapable of defining a topological boundary on its own; it can only link existing arguments/events. Therefore, to construct an absence, the grammar cannot rely on a simple preposition plus a negative feature. It is forced to project specific nominal or spatial lexical material — an Axial Part (AxP)/Relational Noun — that serves exactly as the "internal skin" or interface of the empty region. By projecting an AxP, the syntax turns a negative concept (an absence) into a positive structural object (a boundary) that can be grammatically manipulated.

Crucially, the exact same structural alternation between analytic and grammaticalized markers observed for exceptives in (6) is found in the caritive domain. The analytic abessive locution *in assenza di* ('in absence of', also *in mancanza di*, 'in lack of') in (12a) makes the underlying bipartite DRM structure fully transparent, with both the upper relator (*in*) and the lower relator (*di*) overtly filled.⁶

(12) Italian

- a. In assenza di aiuto, non parto.
 In absence of help, not leave.1SG
 'Without help, I won't leave.'

⁵ In the realm of formal ontology, plain mereology (the theory of parts and wholes, governed by partial ordering and parthood predicates) is inherently insufficient to account for individual integrity, boundaries, and holes. An *annulus* (a ring shape), for example, contains a central hole. Pure mereology cannot distinguish between a geometric square that is a mereological *part* of the solid annulus and a square that is merely *spatially included* within the central void of the annulus. Topology is absolutely required to account for the spatial relation of something being "inside," "outside," or "surrounding" a void. We argue that the grammar "grants ontological dignity" to the empty region by projecting a boundary, it is effectively mapping topological primitives onto syntactic structures. In mereotopological frameworks, a boundary is not an independent set of points; it is a dependent entity defined through mereotopological interaction, utilizing primitives such as interior parthood and overlapping. A boundary is a "dependent thing" that can only exist as the boundary of something else. This dependency perfectly mirrors the syntactic reality of the Double-Relator Model. The Axial Part is a dependent nominal entity; it cannot project referential independence but must be structurally anchored to a Ground via the lower relator, creating the boundary that delineates the void.

⁶ A further confirmation of the existence of the upper relational node comes from Italo-Romance dialectology. As documented by Rohlfs (1966: 235), in Tuscan varieties (e.g. Montale Pistoiese dialect), the caritive preposition is lexicalized in the composite form *insenza*, lexicalizing the adposition *in*. The morphological emergence of this variant demonstrates the activation of the upper relator in these varieties, the item *in-* acts precisely as a spatial anchor for the Axial Part *senza*, transparently instantiating the [p_upper [AxP]] geometry predicted by the Double-Relator Model.

- b. [p_upper *in* [AxP *assenza* [p_lower *di* [DPGround *aiuto*]]]]
- c. Senza (di) te, non parto.
without (of) you.SG not leave.1SG
'Without you, I won't leave.'
- d. [p_upper $\emptyset$ [AxP *senza* [p_lower (*di*) [DPGround *te*]]]]

When the grammaticalized caritive marker *senza* ('without') is used, as in (12b), it acts as the Axial Part defining the empty region. The upper relator is either a silent Kaynian head ($\emptyset$) or has been absorbed by *senza* via phrasal spell-out (cf. fn. 2). Furthermore, the morphosyntactic behavior of the Ground perfectly confirms the AxP status of *senza*. When the Ground is a strong pronoun, modern Italian allows an optional realization of the lower relator *di* ('of'), yielding both *senza te* and *senza di te*. This optionality is a hallmark of Axial Parts in Italian spatial and temporal grammar (e.g., *dopo il lavoro* 'after the work' vs. *dopo di te* 'after you'; *sotto al tavolo* 'under the table', *sotto di te* 'below you', cf. Franco & Schirato 2026), also reflecting inter-speaker variability in the overt spell-out of *p lower*. True simple prepositions never allow this lower relational layer (**con di te* 'with of you' is strictly ungrammatical). This proves that the caritive marker is never a simple preposition, but a specialized nominal/axial core strictly requiring a bipartite relational structure to delimit the topology of absence.⁷

4. The Semantics of the Ground: Kinds, Possession, and Existential Negation

A purely morphosyntactic and mereotopological analysis of caritives, such as the one provided by the Double-Relator Model (DRM), must interface with the semantic properties of these constructions. In recent formal semantics, two incursions into the domain of caritives and privatives are Matushansky (2019) and Phillips (2022). While both authors offer crucial

⁷ This syntactic reality is supported by perceptual phenomenology (cf. Varzi 2007). Human cognition does not merely register a passive "failure to perceive" when confronted with missing entities; human beings actively perceive absences. Whether it is seeing a fish-shaped hole cut into a piece of paper, noticing the sudden absence of a colleague in a meeting room, or perceiving a gap on a bookshelf, the cognitive experience is one of positive apprehension of a negative space. Crucially, absences are ontologically dependent entities; an absence is always the absence *of* something specific, and it is usually defined relative to an expectation of presence. Syntactically, the Axial Part reifies this cognitive absence into a structural node, and the lower relator links that absence to the specific missing Ground, fulfilling the ontological dependency requirement. The grammar does not compute nothingness; it computes a bounded region characterized by the specific exclusion of the Ground.

insights into the logical interpretation of absence, they leave the syntactic derivation largely underspecified. In what follows, we will demonstrate that their semantic findings not only are perfectly compatible with the DRM, but actively require a bipartite relational architecture $[p_upper [AxP [p_lower Ground]]]$ to be properly derived at the syntax-semantics interface.

4.1 Matushansky (2019): Relational Kinds and the Lower Relator (*p lower*)

Matushansky (2019) investigates the scopal properties of caritive prepositions like French *sans* or English *without*. She observes a fundamental paradox when trying to compute the semantics of incorporation and negation in seemingly simple caritive phrases containing bare nouns, such as (13).

- (13) French (Matushansky 2019: 6)
- | | | | |
|---------------------------|-------|---------|---------|
| une | porte | sans | poignée |
| a | door | without | handle |
| 'a door without a handle' | | | |

In these cases, Matushansky says that if we treat *sans* purely as a negative operator taking an existential quantifier over individuals, the standard compositional options both fail. As Matushansky points out, evaluating the existential over negation

$(\lambda x.\text{door}(x) \wedge \exists y[\text{handle}(y) \wedge \neg \text{HAVE}(x,y)])$ is trivially true, because there is always *some* handle in the world that a specific door does not have. Conversely, evaluating negation over the existential $(\lambda x.\text{door}(x) \wedge \neg \exists y[\text{handle}(y) \wedge \text{HAVE}(x,y)])$ asserts a total lack of handles, but clashes with the bare noun, which should presuppose the unicity of the handle.

Matushansky's solution is that the bare nominal Ground in caritives does not denote properties of individuals, but rather *concepts* or *relational kinds*. Crucially, she argues that the caritive relation is entirely parasitic on an unspecified, contextually salient possessive relation. The caritive asserts that the concept (*the handle*) is not instantiated on the possessor (*the door*).

A potential issue with this purely semantic account is that it forces the lexicon to stipulate *without/sans* as a highly complex functional type (e.g., $\langle e, \langle \langle e,t \rangle, \langle e,t \rangle \rangle \rangle$) that accesses an invisible, non-syntactic possessive relation.

The DRM resolves this issue by providing the exact structural skeleton for Matushansky's semantic insight. In our model, the caritive marker is not a preposition, but an Axial Part (AxP) that requires a lower relator (*p lower*) to license its Ground. The *p lower* is precisely the

elementary predicate of inclusion ($\subseteq$) that inherently encodes possession (Franco & Manzini 2017).

(14) [p_upper $\emptyset$ [AxP *sans* [p_lower $\emptyset(\subseteq)$ [NP_{Concept} *poignée*]]]]

In (14), the semantic need for a possessive link is not contextually stipulated, but syntactically inherent: the null lower relator ($\subseteq$) structurally relates the concept-denoting NP to the empty topological region defined by the AxP. When the Ground intrinsically lacks a full D-layer (as bare concept nouns do), it merges directly with the (silent) *p_lower*. However, when the Ground is a referential individual (a DP or a strong pronoun), the lower possessive relator must be overtly spelled out, which is exactly why languages like Italian insert the genitive preposition *di* ('of') to mediate the relation (e.g., *senza di te* 'without of you'). Thus, the DRM proves that the possessive *parasitism* of caritives is a direct reflex of their bipartite syntactic projection.⁸

4.2 Phillips (2022): Negative Existentials, Indexicality, and the Upper Relator (*p_upper*)

Another theoretical contribution comes from Phillips (2022), who investigates privative / abessive markers in Australian (Pama-Nyungan) languages. Phillips demonstrates that privative cases in these languages do not merely modify nouns, but function as fully-fledged *negative existentials*.

Semantically, building on Francez's (2007) quantificational treatment of existential predication, Phillips defines a privative (PRIV) marker as a generalized quantifier — an operator $\lambda P \lambda Q. \text{no}(P, Q)$ that asserts an empty intersection between a "Pivot" (the entity whose existence is under discussion, *P*) and a "Coda" (the contextual domain, *Q*).

Consider the data from Muruwari. In this language, the introduction of a new referent into the discourse (an affirmative existential) is encoded precisely by a comitative / spatial phrase (*-yita* 'with'), whereas the negative existential requires the dedicated privative suffix *-kil*:

⁸ Within the DRM architecture, when the Ground intrinsically lacks a full determiner layer (as bare concept nouns do), it merges directly with a phonologically silent lower relator. This highly specific structural configuration feeds the semantic need for an instantiation link between the concept-denoting Noun Phrase (the kind) and the empty topological region defined by the Axial Part. By treating the Axial Part as the boundary of a void, the syntax dictates that the *kind itself* is excluded from the spatial or eventive zone of the Figure. The grammar is not quantifying over individual *handles* in the physical world; it is excluding the abstract concept of a *handle* from the relevant topological zone of the *door*.

- (15) Muruwari
 thuu kuya-yita wartu
 much fish-COM hole.ABS
 'The river has a lot of fish in it.'
 (lit. The hole (is) with much fish)

- (16) Muruwari
 palanj mathan-kil
 nothing stick-PRIV
 'There are no sticks.'

For Phillips, the privative *-kil* in (16) asserts that there is nothing in the intersection of the set of *sticks* (the Pivot) and the contextual domain (the Coda, *Q*). While Phillips' semantic characterization of privatives as negative existentials is elegant, treating PRIV simply as a primitive logical quantifier (*no*) obscures its morphosyntactic nature. If PRIV is just a boolean negative operator, why is it so morphologically heavy and distinct from standard clausal negation? Furthermore, why do expressions of existence and non-existence systematically draw from the spatial domain, as seen with the Muruwari comitative in (15)?

Here, a crucial idea regarding the nature of existential constructions provides the missing link to the DRM. Existential propositions exhibit "radical context dependence" and bear a strong functional connection to notions of indexicality and spatial location. In Francez's (2007) semantic terms, an existential proposition like *There are no trees* cannot be evaluated in a vacuum; it requires a "contextual domain" (often the spatio-temporal conditions of the utterance). Syntactically, as argued by Franco & Lorusso (2018), the expression of existence is inherently tied to elementary predicates that structurally anchor the event to a specific domain. As said above, pure mereology or abstract Boolean logic cannot compute an absence without a topological space to anchor it (Varzi 2007). To compute the empty intersection $no(P,Q)$, natural language syntax cannot simply merge a NEG head; it must project a topological boundary that roots the absence in a specific deictic or contextual space. This deep connection between spatiality, existentials, and absence is strikingly confirmed by Amazonian languages. Guillaume (2022) demonstrates that in Tacana (a Takanan language of Bolivia), the standard negator *aimue* actually originates from a negative existential predicate meaning 'be without' or 'non-existent'. When asserting absence, Tacana does not use a clausal negative operator, but directly merges this specialized existential item.

(17) Tacana

Kwati=mu aimue=tsu'u

firewood=CNTR nonexistent=still

'There is no firewood yet.' (lit. firewood was nonexistent)

Crucially, affirmative existentials in Tacana are expressed via highly specified spatial posture verbs (e.g., *ani* 'sit', *neti* 'stand', *sa* 'lie'), confirming that the grammar conceptualizes existence as a topological region. When existence is negated, the specific spatial posture is often removed, and the dedicated element *aimue* steps in to bound the empty region.

(18) Tacana

[Da tiempo] aimue sapato ani-ina=mawe.

that time NEG shoe sit-HAB.PST=NEG

'At that time, there were no shoes.'

The privative/negative existential markers — whether the Muruwari *-kil* or the Tacana *aimue* — are exactly the Axial Parts (AxPs) projected by the grammar to give "ontological dignity" to this spatial/contextual void.

(19) [p_upper ∅ [AxP *-kil* / *aimue* [p_lower ∅ [NP *mathan* / *sapato*]]]]

Under the DRM, Phillips' semantic "Pivot" (the *stick* or the *shoe*) maps directly onto the internal argument of the lower relator (*p lower*), which establishes the conceptual expectation of presence. The semantic "Coda" (the contextual domain, such as [*da tiempo*] in Tacana) maps onto the external argument introduced by the upper relator (*p upper*), realizing the mereological anchoring discussed by Franco & Lorusso (2018) for Romance-Based Creoles. The AxP acts as the 'boundary', effectively grammaticalizing the "negative existential" space. This strictly phrasal approach elegantly explains the Domain Expansion of privatives into standard negation (the Negative Existential Cycle, see Croft 1991).⁹ As privatives

⁹ The Negative Existential Cycle (cf. Veselinova & Hamari 2022) is a well-documented grammaticalization pathway where languages recruit a "special" negative existential predicate (a verb or adjective meaning "not exist," "lack," or "be empty") which eventually displaces older negators and expands its domain to become a standard clausal negative marker or a privative/caritive adposition. Negative existentials develop cross-linguistically from lexical sources with meanings inherently related to spatial voids, such as "lack," "absent,"

grammaticalize, their scope expands from individuals to *eventualities*. If the AxP is an empty boundary, its scope depends entirely on the nature of the *p upper* relator. When *p upper* attaches to a nominal host, the AxP creates a nominal privative (a *stick-less* space). When *p upper* attaches to a *vP* or an eventive nominal (as in the grammaticalization of Tacana *aimue* into a clausal negator), the AxP bounds an event, creating the absence of a co-location relation. The syntax of the negative existential cycle is therefore not a semantic shift from 'without' to 'not', but a simple reflex of the syntactic attachment height of the [p_upper [AxP]] complex within the clausal spine, operating on different contextual domains.

To summarize, the cross-linguistic landscape of privation is highly complex, and formal approaches have often targeted different stages of its grammaticalization. Phillips (2022) demonstrates for Pama-Nyungan languages, nominal "privative" suffixes frequently act as negative existential predicates that can eventually grammaticalize into standard clausal negators (following the Negative Existential Cycle). While these dynamically shifting Australian structures are not strictly congruent with the stable, adpositional abessives of Romance and Germanic (like *senza* or *without*), Phillips's core semantic insight is crucial for the "big picture". He formally argues that privation fundamentally predicates the *absence of an entity within a restricted spatial/contextual domain*, rather than operating as a basic boolean operator. This aligns with our morphosyntactic definition of caritives as projecting a boundary of "zonal absence".

5. Cross-Linguistic data

The fundamentally phrasal and complex nature of caritive constructions proposed by the Double-Relator Model finds robust empirical support in cross-linguistic typology. In their study on a sample of 239 languages, Stolz, Stroh & Urdze (2007) demonstrate that the abessive (caritive) is never merely the symmetrical, negative equivalent of the comitative or instrumental. Rather, the expression of "absence" systematically behaves as a highly marked category, regularly exhibiting greater morphological, segmental, and syntactic complexity than its positive counterparts.¹⁰

"empty," or "destroy" (e.g., Turkish *yok*, Kwangali *tupu* "empty"). Because negative existentials originate as full lexical items (verbs or adjectives of absence), they naturally map onto the Relational Noun/Axial Part node in the Double-Relator Model. Thus, the grammar does not invent a preposition of absence; it recruits a lexical noun or verb of emptiness and grammaticalizes it into the AxP position.

¹⁰ Stolz et al. (2007) show that in over 80% of segmentally unequal pairs encoding presence versus absence, the abessive contains more phonological segments than the comitative. These data support the idea that the

This cross-linguistic markedness elegantly mirrors the architectural predictions of the DRM: while standard inclusion (comitative/instrumental) can often be conveyed by a simple primitive relator ($\subseteq$ / $\supseteq$), "zonal exclusion" or privation requires the complex, bipartite geometry of an Axial Part combined with a lower relator (*plower*). The difficulty that natural language syntax faces in encoding absence is evident in the typological distribution of caritives. In many languages, a dedicated adposition for "without" simply does not exist. Stolz et al. (2007) identify two major cross-linguistic strategies to compensate for this gap. The first (Type A) relies on the direct negation of the inflectional comitative/instrumental, as seen in Hixkaryana (e.g., *y-akoro-hra*, '3SG-with-NEG'). The second strategy (Type B) is even more telling: languages resort to full bi-clausal structures or serial verbs involving negative existentials or predicates of non-use. In Japanese, for instance, the negative instrumental is encoded by the negated verb *tsukawazu* ('not using'), while the negative comitative is expressed by *tsuretai* ('not taking/bringing').

In the most comprehensive cross-linguistic surveys of these categories to date, Stolz, Stroh & Urdze (2006, 2007) demonstrate that the world's languages can be classified based on their encoding of comitatives and instrumentals. For positive relations, they identify three language types: Type A (which formally distinguishes comitatives and instrumentals), Type B (typical of Standard Average European, which uses a single syncretic marker), and Type C (which presents a mixed system).

However, when shifting from the grammar of presence to the grammar of absence, this symmetric typology breaks down into a profound markedness asymmetry. Stolz et al. (2007) observe that the abessive is systematically more complex. Languages resolve the encoding of absence through three major structural strategies (labeled Type A, Type B, and Type C). Strikingly, all three typological strategies can be elegantly captured by the DRM as specific syntactic operations computing exclusion.

In Stolz et al.'s (2007) typology, Type A languages (such as Hixkaryana) are described as lacking a dedicated, free-standing grammaticalized marker for the abessive. Instead, these languages encode the abessive by directly attaching a "negative" morpheme to the comitative/instrumental marker.

projection of an Axial Part to define a spatial void requires more phonological material and demands greater syntactic independence (free morpheme status) than a primitive, unmarked inclusion relation.

(20) Hixkaryana (Stolz et al. 2007: 71) =(1b)

kamo ro-akoro-hra sun 1SG-with-NEG

'the sun without me'

Under a superficial view, Type A languages might seem to challenge the DRM: if absence can be expressed simply by adding a boolean negative operator to an inclusion predicate ($\supseteq$ + NEG), why postulate the necessity of an Axial Part? From a theoretical standpoint, two cautious analytical pathways arise within our framework to account for this type.

It is entirely possible that the affix in certain Type A languages operates as a genuine boolean negator. However, if this is the case, the DRM predicts that the resulting structure is *not* a true phrasal caritive. Just as boolean coordination (*I saw all mammals, but not penguins*) differs fundamentally from a true exceptive (*# I saw all mammals except penguins*), a boolean negation over a comitative merely denies the truth of a co-occurrence at the eventive or clausal level (yielding a reading akin to *and/but not with*). The syntax is not projecting a "zone of absence" modifying a noun, but simply negating a secondary predication.

Alternatively, it is possible to propose that in many cases, the affixes labeled broadly as "negative" by descriptive typologies actually function as spatial or directional excluders. Striking empirical evidence for this comes from the internal anatomy of the English caritive marker *without*. While English is not an agglutinative language, its abessive marker is transparently composed of the comitative preposition *with* and the purely spatial marker *out*.

(21) [p_upper with [AxP out [p_lower $\emptyset$ [DP you]]]] (*English: without you*)

We assume that in (21), *out* is a spatial Axial Part (a boundary-defining element). The upper relator (*p_upper*) is overtly realized by *with*, while the lower relator (*p_lower*) remains phonologically silent ($\emptyset$), satisfying the c-selection of the Ground. This demonstrates that even when a language derives an abessive directly from a comitative base, it does not necessarily attach a logical boolean negator; rather, it can merge a spatial element (like *out* or the suffix *-less*) to bound the empty region. Thus, Type A "agglutination" may mask a transparent morphological spell-out of the [p_upper [AxP]] complex spine.

Type B languages in Stolz et al.'s (2007) classification (e.g., Dravidian, Oceanic, and Australian languages) similarly lack a dedicated adpositional or affixal abessive marker. Instead, they

express caritive semantics through biclausal constructions or by employing a full negative-existential clause.

This typological class provides striking macro-areal support for the analysis developed in Section 4.2 based on Phillips (2022). As established in section 4.2, an existential absence requires a contextual spatial domain. If a language lacks a dedicated nominal Axial Part to act as the boundary of the void at the phrasal level, the grammar must resort to a higher functional structure to anchor the absence. It projects an entire eventive shell (a negative existential predicate) where the Ground is evaluated against the contextual domain. Type B languages prove that the topological boundary required to compute an absence must be syntactically projected; if it is not available in the adpositional lexicon, it is forced upward into the clausal spine.

Type C, as we have seen for Italian, represents languages that possess distinct, dedicated, and highly grammaticalized expressions for the abessive. This is the hallmark of Indo-European and Uralic languages. The Persian language provides striking overt evidence for the structural complexity predicted by the DRM for Type C markers.

5.1 *Abessives and the Internal Anatomy of Axial Parts: Persian*

In Persian, the comitative relation is expressed by the simple monomorphemic preposition *ba* ('with', inverse inclusion $\supseteq$). However, the caritive resorts to the marker *bedun-e* ('without'). Following the syntactic analysis of Persian Class 2 prepositions as Axial Parts (Pantcheva 2006) and the morphosyntactic mapping of the DRM, the marker *bedun-e* is demonstrably multi-morphemic. It spells out every node of the [p_upper [AxP [p_lower Ground]]] structure. Historically and synchronically, *bedun-e* decomposes into three distinct functional elements: a) *be-*: the standard Persian dative/allative preposition ('to/at'). In the DRM, this functions as the *pupper* relator; b) *-dun-*: The nominal root acting as the Axial Part (AxP), creating the topological boundary; -e: The Ezafe particle. The *Ezafe* acts exactly as the *p lower* D-linker (see Franco et al. 2015), mediating the selectional relationship to the Ground.

- (22) [p_upper *be* [AxP *dun* [p_lower *-e* [DP_{Ground} *kilid*]]]]
(Persian: *be-dun-e kilid*, 'without a key')

The Persian data prove that grammaticalizing a Type C absence inherently requires the construction of a spatial/mereological complex PP.

Note that in Persian, "absence" can be morphologically encoded via the element *bi-*. Crucially, *bi-* is not a preposition on par with *be* ('to') or *ba* ('with'); rather, it is a derivational prefix that operates strictly in the lexicon to derive nouns and adjectives. It denotes an intrinsic, stable, or characterizing property of absence. For example, it generates forms such as *bi-pul* ('without money' → 'poor'), *bi-fāyde* ('without use' → 'useless'), and *bi-ehsās* ('without feeling' → 'insensitive'). In these instances, absence is treated as a permanent quality of the entity, not as a relational circumstance of an event (cf. a set of Italian compounds formed, with *senza*, as for instance *senzatetto* 'homeless') (cf. Mahootian 1997).

Conversely, when absence is framed as a situational or circumstantial relation modifying an event (true caritive syntax), Persian employs the preposition *be-dun*. The contrast between the comitative/instrumental and the caritive in Persian perfectly encapsulates the markedness of absence discussed by Stolz et al. (2007: 78-79). The comitative *ba* ('with') is a simple relator: it is monosyllabic and monomorphemic (a simple CV skeleton). The caritive *bedun-e*, however, exhibits significant segmental and morphological complexity.

Most importantly for the DRM, *bedun* functions categorically as an Axial Part (or a Relational Noun). It cannot merge directly with its Ground; rather, it obligatorily requires the overt realization of the lower relator, surfacing as the *Ezafe* particle (-*e*). Thus, the Persian data converges with the Italian evidence: true caritives are not derived from clausal ellipsis, but project a complex adpositional structure consisting of an Axial Part defining the exclusion zone, anchored to its nominal Ground via a lower genitive-like linker (the *Ezafe* in Persian).

5.2 The Slavic Pattern

The requirement for a lower relational layer (*plower*) in Type C languages is not restricted to overt linker particles like the Persian *Ezafe*. In heavily inflected languages, this lower relational layer surfaces as Genitive case, the universal morphological reflex of the adnominal part-whole inclusion predicate ($\subseteq$). Slavic languages offer a textbook illustration. In Russian, the comitative relation is encoded by the preposition *s/z* ('with'), which systematically assigns Instrumental case to its complement—an event-anchored oblique case perfectly suited for direct integration via the $\supseteq$ predicate (see Stolz et al. 2007).

(23) Russian

- a. *s* *brat-om*
 with brother-INSTR
 'with the brother' (Comitative)

- b. *bez* *brat-a*
 without brother-GEN
 'without the brother' (Caritive)

Conversely, the abessive preposition *bez* ('without') strictly assigns Genitive case (23b). Why should the marker of absence require the genitive case, typically reserved for adnominal possession? Under the DRM, this is a structural necessity. Because the caritive marker *bez* is fundamentally an Axial Part defining an empty spatial region, it cannot directly assign an event-related case like the Instrumental. As a noun, it must link to its Ground via the nominal inclusion predicate ($\subseteq$). The Genitive case suffix (-a) on the Ground is exactly the morphological spell-out of this *plower* relator.

(24) [p_upper $\emptyset$ [AxP *bez* [p_lower -a (GEN) [DPGround *brat*]]]]

The Slavic Genitive assignment under *bez* is thus the exact morphosyntactic equivalent of the Persian *Ezafe* under *bedun* and the Italian preposition *di* under *senza*. They are all overt manifestations of the lower relational head required to license the nominal boundary of a topological void.

5.3 The Bantu Locative Classes as Topological Boundaries

A potential challenge to the mereo-topological necessity of Axial Parts comes from languages with highly grammaticalized, agglutinative negation systems, such as the Bantu family. In many Bantu languages, standard negation is expressed via bound inflectional prefixes (e.g., pre-initial *ha-* or post-initial *-si-*). If such a language were to express the caritive simply by attaching a negative prefix to a comitative preposition, one might argue that the grammar is computing absence via a pure, floating boolean operator, without projecting any topological boundary.

However, a closer look at the morphosyntax of Bantu reveals that this is never the case. First, the standard negation affixes themselves are not abstract boolean operators. Extensive diachronic typological work (Güldemann 1999; Nurse 2008) demonstrates that post-initial negators in Bantu grammaticalized from cessative auxiliary verbs, such as Proto-Bantu **dèk* ('to let go, leave, cease') or Manda *-kotok-* ('to leave off, stop') (Bernander, Devos & Gibson 2022). Thus, even at the clausal level, negation in Bantu operates as an eventive or spatial excluder.

More importantly, when Bantu languages compute a zonal absence (a caritive or negative existential relation), the syntax strictly prohibits the direct negation of the comitative relator. As extensively documented by Bernander et al. (2022), the encoding of absence systematically requires the projection of Locative Class markers (specifically, for Swahili, Class 16 *pa-*, Class 17 *ku-*, or Class 18 *mu-*).

Consider the Swahili caritive/existential negator *ha-pa-na* 'without / there is not'. It is morphologically composed (only) of the standard negative prefix (*ha-*), the Class 16 spatial locative marker (*-pa-*), and the comitative relator (*-na* 'with').

(25) Swahili (Bernander et al. 2022: 71)

ha-pa-na	m-tu
NEG-LOC.16-COM	1-person
'without a person / there is no person'	

The presence of the locative class marker *-pa-* is crucial. It proves that the grammar cannot simply output a [NEG [with [DP]]] structure. To compute the absence, the syntax must project a spatial domain (the locative class 16) that acts exactly as the Axial Part in the DRM. The string *ha-pa-na* translates literally to 'the spatial region is not with [the person]'.
Furthermore, another widespread strategy for negative concomitance across the Bantu domain involves reflexes of the Proto-Bantu nominal stem *-tópú*, which translates literally as 'empty space' (Bastin et al. 2002; Bernander et al. 2022: 75). In languages like Kwangali or Ndengeleko, this root is combined with a locative prefix to express the absence of an entity.

(26) Ndengeleko (Ström 2013, apud Bernander et al. 2022: 75)

n-tópú	oomba
LOC.18-empty.space	10.fish
'without fish / there is no fish'	

The Bantu data provide further cross-linguistic evidence for our model. They demonstrate that even in highly agglutinative languages lacking independent adpositional AxPs like *senza* or *without*, the syntax is forced to resort to locative class morphology or nouns meaning 'empty space' (*-tópú*) to give 'ontological dignity' to the void. The topological boundary required by Varzi (2007) is thus morphologically reified as a locative class projection.

5.4 Clausal Negative Concomitance: Unifying Phrasal and Clausal Caritives

An exhaustive theory of the grammar of absence must also account for clausal negative concomitance. Commonly referred to in the typological literature as "without V-ing" clauses (e.g., *He slept without using a pillow*), these constructions present a highly complex cross-linguistic interaction between clause-linkers and standard clausal negative markers. If the Double-Relator Model (DRM) posits that caritive meaning is inherently relational and spatial (zonal exclusion) rather than a product of boolean clausal negation, how does it account for the overt presence of standard negative markers in the clausal caritives of many languages? A recent cross-linguistic survey by Olguín Martínez & Peregrina Llanes (2023) demonstrates that the behavior of standard negative markers within dependent caritive clauses varies predictably. Languages are divided into those using *monofunctional* (dedicated) versus *polyfunctional* (generic) linking devices. This typological split perfectly aligns with the architectural predictions of the DRM.

When a language utilizes a dedicated, highly grammaticalized caritive linker whose sole purpose is to encode absence (a monofunctional device), the addition of a standard negative marker inside the dependent clause is strictly disallowed or conceptually redundant. This robustly supports the DRM's core thesis: the dedicated marker occupies the Axial Part node, which natively provides the absolute boundary of zonal exclusion, rendering any internal boolean negation structurally superfluous. Consider the Persian example in (27):

- (27) Persian (Olguín Martínez & Peregrina Llanes 2023: 56)
- | | | | | |
|------------|------|-----------|---------------------|--------|
| bedun-e | inke | laebxaend | be-zaen-aed | raeft |
| without-EZ | that | smile | PRS.SUBJ-strike-3SG | go.PST |
- 'He left without smiling.'

Crucially, the DRM elegantly explains how a relational Axial Part, which natively C-selects for a nominal entity, can embed a clausal Ground. In monofunctional clausal caritives like Italian *senza che*, Spanish *sin que*, or Persian *bedun-e inke*, the element *che/que/inke* does not function as a standard clausal complementizer projecting a phantom CP. Drawing on extensive evidence that Romance complementizers of the *wh*- series are intrinsically nominal elements (Manzini & Savoia 2003, 2011), we argue that *che* operates as a generalized D-linker. This D-linker "dresses" the fully projected clause as a nominal entity, thereby perfectly satisfying the strict categorial selection requirements of the Axial Part's lower relator

(*plower*). This mechanism perfectly unifies the syntax of phrasal and clausal caritives under the exact same (mereotopological) DRM model.¹¹

Conversely, when a language lacks a dedicated, monofunctional Axial Part for absence at the clausal level, it is forced to use a general, polyfunctional clause-linker (e.g., a simultaneous converb typically meaning "while" or "when"). In these structures, standard negative markers are mandatory to force the caritive reading. For instance, in Sidaama, the converb *-nni* acts generally as a "while" operator when the clause has positive polarity; to express "without V-ing," the standard negative marker *-kki* must obligatorily attach to the embedded verb:

(28) Sidaama (Afro-Asiatic) (Kawachi 2007: 382)

keeš-i-tto-kki-nni	amo.
stay.long-PFV-2SG.M-NEG-CVB	come.IMP.2SG '
Come without staying long.'	

Rather than contradicting the Double-Relator Model, this typological bifurcation precisely confirms its architectural boundaries. The DRM models the syntax of adpositional/relational structures, where a nominal AxP intrinsically provides the topological boundary of zonal exclusion, rendering boolean negation structurally illicit. However, polyfunctional linkers like the Sidaama converb *-nni* do not project an adpositional/nominal structure. A converb is a verbal affix that projects a non-finite clausal adjunct.

In other words, constructions utilizing polyfunctional linkers represent an entirely different syntactic strategy. Because the language lacks a lexical item to instantiate the boundary-defining Axial Part node, it cannot compute "absence" topologically via relational syntax. Instead, the grammar is forced to resort to standard clausal adjunction. Within this purely verbal domain, the only available mechanism to derive the caritive meaning is compositional:

¹¹ The structural uniformity between phrasal and clausal caritives (as well as temporal constructions like *prima che* 'before that' and *dopo che* 'after that') is theoretically grounded in the reanalysis of Romance complementizers. Standard approaches assume that *che/que* instantiates a verbal functional head within the clausal left periphery (e.g., C or Force). However, Manzini & Savoia (2003, 2011) extensively demonstrate that Romance *che* is intrinsically a nominal head (a D-element belonging to the *wh*-argumental series) which takes the embedded sentence as its complement. This view strongly converges with the later framework of Kayne (2010), who explicitly argues against *che/that* as a Force head or a CP phase head, reanalyzing it instead as a relative pronoun — hence, a nominal element — that mediates sentential complementation. Under this unified nominal perspective, when adpositional Axial Parts natively selecting for a D-layer embed a finite clause (e.g., *senza che potessi salutarlo* 'without that I could greet him', or *prima che potessi salutarlo* 'before that I could greet him'), the element *che* does not project a phantom verbal CP. Instead, it functions as a generalized D-linker, providing the necessary nominal "dressing" to satisfy the strict c-selectional requirements of the Axial Part's lower relator.

the syntax must employ the standard boolean clausal negator (*-kki*) to negate the eventive predicate inside the temporal adjunct. The Sidaama construction therefore literally translates to a temporal adjunct containing a negated event: *Come [while [NOT staying long]]*.

In sum, the DRM predicts the cross-linguistic distribution of negative markers in clausal concomitance. Zonal exclusion has an absolute structural cost: if it is resolved via nominal/relational syntax, a dedicated Axial Part is merged (with a D-linker) and boolean negation is banned. If the grammar lacks the appropriate AxP, it must fall back on verbal/clausal syntax (e.g., polyfunctional converbs), where boolean negation becomes absolutely mandatory to compositionally ‘simulate’ the absence.

6. Discussion and Conclusion

This paper set out to solve a long-standing typological and theoretical puzzle: why is the grammatical encoding of absence (the caritive/abessive) systematically more complex, morphologically heavier, and less bound than the encoding of presence (the comitative/instrumental)? By integrating formal syntax, semantics, and mereotopology, we have argued that this asymmetry is not an accidental tendency of natural language, but a structural necessity. Comitative and instrumental relations express straightforward inverse inclusion ($\supseteq$) and can therefore be directly lexicalized by simple, monomorphemic elementary predicates (e.g., *con*, *with*, or simple bound affixes). A caritive relation, however, asserts a *zonal absence*.

Drawing on Varzi’s (2007) mereotopological framework, we have demonstrated that pure mereology cannot compute a void or a “hole”. To be grammatically manipulated, an absence must be granted “ontological dignity”, which requires tracing a discrete topological boundary. Syntactically, a simple prepositional head is incapable of defining a boundary; therefore, the grammar is forced to project specialized spatial or relational nominals—namely, Axial Parts (AxPs).

The Double-Relator Model (DRM) captures this necessity, predicting a bipartite configuration [p_upper [AxP [p_lower Ground]]]. This layered architecture is the source of the morphological weight and typological markedness of abessives cross-linguistically. Furthermore, it elegantly accommodates the semantic properties of caritives: the lower relator (*p_lower*) satisfies the inherent possessive relation required by kind-denoting remnants (Matushansky 2019), while the upper relator (*p_upper*) can anchor the negative

existential boundary (the AxP) to a specific spatial or eventive contextual domain (Phillips 2022).

The analysis proposed here has broader implications for our understanding of the syntax-semantics interface and the grammaticalization of logical operators.

First, the data strongly suggest that natural language resists encoding phrasal "absence" through pure floating boolean operators (e.g., standard logical negation). As we have seen in our dismantling of the typological "agglutinative" strategy (Type A), even when an abessive appears to be derived from a negated comitative, the internal anatomy of markers like English *with-out* reveals the obligatory projection of a spatial excluder. We argue that the logic of exclusion at the phrasal level is fundamentally spatial and topological, not strictly boolean. Second, the present work unifies the grammar of the caritive with the grammar of exceptive constructions (e.g., *everyone except John*). Exceptive markers (*tranne, eccetto*) are also Axial Parts sandwiched between relational heads, serving to carve out a boundary. The difference lies solely in the domain of application: while exceptives operate on a quantificational domain (executing a set subtraction, $W \setminus P$), caritives operate on the eventive or possessive spine (executing a zonal exclusion). The structural repair mechanism—the projection of an AxP to limit the void—is mathematically and syntactically identical.

Finally, the overt realization of the complex DRM structure in Type C languages—such as the Persian *be-dun-e* (dative *pupper* + AxP + *Ezafe plower*) or the systematic Genitive case assignment triggered by the Slavic abessive *bez*—confirms that complex prepositions and oblique cases are not opaque, unanalyzable idioms. They are fully transparent syntactic workspaces where elementary predicates compute inclusion, location, and boundaries. To conclude, the grammar of absence is inherently heavier because computing "nothingness" is a derived, topologically complex operation. By mapping the philosophical concept of a topological boundary onto the syntactic projection of Axial Parts via the Double-Relator Model, we obtain a principled, universal architecture that seamlessly aligns the semantic interpretation of negative existentials with the robust cross-linguistic markedness of caritive constructions.

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